The Wrong Unit of Uncertainty: Adaptive Design-Aware Conformal Inference for Repeated Cross-National Surveys

The Authors

Abstract

Comparative research routinely reads a single wave-pair mean contrast and narrates it as a persistent, distribution-wide, country-level trend—attaching uncertainty to the wrong unit and over-detecting change. We give a principled instrument: a finite-sample simultaneous band over a country's whole attitude *trajectory*, with the country as the exchangeable unit, and a hierarchy of claims (pairwise → any-pair → net → persistent country-wide) whose top rung is the honest object. A second error compounds it—the calibration objects, other countries' distributions, are not observed but estimated from complex samples—and we prove this is a hard limit: bands calibrated on noisily-estimated objects face a non-identification barrier (Theorem 1) and a distribution-free reliability floor requiring $K \geq 94$ exchangeable populations (Proposition 1). Across the two largest cross-national surveys these barriers bind in opposite ways and never jointly release, so a design-aware correction is provably unreachable at survey scale. What remains is the honest procedure for the regime that occurs: a clustered band exact at any K (Theorem 3) inside a selector that abstains when deconvolution is unreliable, provably correctly. The payoff is two corrections. In the ESS (2018–2022), persistent distribution-wide trust decline is certified in one of roughly thirty democracies (Greece), not the twenty flagged marginally; on the World Values Survey (1981–2022) the marginal case for Foa–Mounk "deconsolidation" over-counts the persistent phenomenon by 2.6–6.5×.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament, satisfaction with democracy, or support for the political system in a country and year is a whole cumulative distribution function over a response scale, and repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey, the AmericasBarometer (Cohen et al., 2023)—let us track how that distribution moves across waves and, increasingly, *transport* it: predict one country's trajectory from the others, or flag which countries depart from a common pattern. Prediction bands and conformal methods are a natural tool for such transport, because they promise finite-sample, distribution-free coverage.

Two wrong units. A claim and its calibration can each be attached to the wrong object. The *claim* "democratic trust is declining" is typically built from a single wave-pair mean contrast, then narrated as a persistent, distribution-wide, country-level trend. Requiring the whole distribution to shift, *simultaneously* across the entire trajectory, is a strictly stronger and more honest object, and—as we find—the number of countries that qualify collapses between the two. Our instrument is a finite-sample simultaneous band over a country's whole trajectory, with the country as the exchangeable unit, and an ordered hierarchy of claims (pairwise → any-pair → net → persistent country-wide $\to$ Bonferroni-across-countries) whose top rung is the object casual readings implicitly assert. This much needs no survey design at all; it is the multiplicity-honest core, and it applies to any repeated distributional claim.

The second, subtler wrong unit is a layer of uncertainty that these exercises silently omit. Conformal inference is usually calibrated on *observed outcomes*: the calibration set is data we actually saw. Repeated cross-national surveys are different. The calibration objects—the other countries' attitude distributions—are themselves *not observed.* Each is estimated from a complex probability sample with weights, clustering (primary sampling units), and stratification, and therefore carries its own sampling error. Standard practice plugs these survey estimates into the calibration step as if they were the truth. The transport band is then honest about variation *across* countries but blind to the fact that each "country" in the calibration set is a noisy estimate of a latent population function.

The two errors are genuinely distinct. Writing $F_{c,r}(t)$ for country c's latent attitude CDF at threshold t in wave r, and $\widetilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\widetilde{F}_{c,r}(t) - \mu_r(t) \;=\; \underbrace{\big(F_{c,r}(t) - \mu_r(t)\big)}_{\text{transport error}} \;+\; \underbrace{\big(\widetilde{F}_{c,r}(t) - F_{c,r}(t)\big)}_{\text{design error}}, \tag{1}$$

where μ_r is the transport center. A conformal band calibrated on the observed $\widetilde{F}$ scores treats the sum as if it were the transport error alone. When the design error is non-negligible the calibrated band is systematically too wide (it inflates the transport spread by the survey noise), and—more subtly—any attempt to correct for this by subtracting the survey noise can make the band too narrow, because the correction is itself estimated from a small number of countries.

Contributions. We make four. *First, a diagnosis and a portable instrument.* We separate two wrong units—the claim unit (a wave-pair contrast standing in for a persistent distribution-wide trajectory) and the calibration unit (estimated distributions treated as latent, the omitted design-error layer of (1))—and give the multiplicity-honest guarantee hierarchy that corrects the first. This much needs no design model and applies to any repeated distributional claim. *Second, and centrally, the boundary of what any design-aware band can do.* An impossibility result (Theorem 1) shows that without a model of the design-noise law the latent transport-error distribution is non-identified—recovering it is a deconvolution problem—so no function of the observed scores is both valid and strictly narrower than the conservative plug-in; the design information (replicate weights, a stratified-PSU bootstrap) is the *identifying* ingredient, not a convenience. We then show that even *with* that information the correction is *unreachable* at cross-national scale, for two structural reasons we make quantitative: the design-to-total noise ratio ρ saturates well below the level at which deconvolution helps, and the deconvolved scale's relative error obeys a distribution-free floor $D \geq \sqrt{2/(K-1)}$ that needs $K \geq 94$ exchangeable populations. On the two largest repeated cross-national surveys these barriers bind in *opposite* ways—the World Values Survey clears the size barrier ($K \approx 100$) but carries only weights-based, negligible design noise, while the ESS has genuine clustered design noise but far too few countries—so neither survey clears both, and the two requirements are structurally anti-correlated across the survey landscape. Design-aware deconvolution is therefore not merely unused but provably unusable here; mapping that boundary precisely is the contribution. *Third, the provably-valid procedure for the regime that actually occurs.* Since the deconvolution is out of reach, the honest deployed object is its reduction: a clustered "population" conformal band that is finite-sample exact at *any* K (Theorem 3), wrapped in a nominal-safe selector that would deconvolve only when four target-blind gates certify reliability and otherwise reduces to the exact band or abstains conservatively (Theorem 5). We prove the abstention is the correct action at survey scale and confirm the frozen procedure on an independent, pre-registered simulation spanning ten data-generating processes it has never seen. *Fourth, two named reanalyses* (Section 8). For European trust (ESS 2018–2022), marginal wave-by-wave estimates suggest widespread decline, but a country-wide, design-aware simultaneous band certifies a *persistent distribution-wide* decline in only one country (Greece). Globally, we reanalyze the Foa–Mounk "democratic deconsolidation" thesis on the World Values Survey (1981–2022) and find its marginal evidence over-counts persistent, distribution-wide deconsolidation by 2.6–6.5×—real in a specific minority, not the broad erosion the marginal reading implies. The lesson generalizes: whenever prediction under shift runs on survey estimates, uncertainty computed for the wrong unit over-detects trends—dramatically for European trust, several-fold for global deconsolidation.

The remainder of the paper sets up the problem and the impossibility result (Section 3), builds the safe adaptive method (Section 4) and its theory (Section 5), validates it in simulation and a design-preserving stress test (Section 6), characterizes the real-data regime (Section 7), reanalyzes political trust (Section 8), and discusses scope (Section 9).

2 Related work

Our work sits at the intersection of four literatures. First, **distribution-free conformal prediction** gives finite-sample coverage without modeling assumptions (Vovk et al., 2005; Lei et al., 2018). Because political data rarely satisfy exchangeability, we build on its relaxations: weighted conformal prediction for covariate shift (Tibshirani et al., 2019) and nonexchangeable conformal prediction, which bounds the coverage gap by the total variation between exchanged distributions (Barber et al., 2023). Gibbs and Candès (2021); Gibbs et al. (2024) adapt conformal sets to distribution shift and interpolate between marginal and conditional validity. Our "adaptive" selector differs in kind: it adapts not to temporal shift but to the *certified magnitude of survey design noise*, reducing to a fixed clustered method when that noise is negligible.

Second, we treat the **population (country) as the exchangeable unit**, following hierarchical conformal prediction for two-layer models (Dunn et al., 2023) and distribution-free inference with hierarchical data (Lee et al., 2023). These methods transport guarantees to an unseen cluster—our cross-national setting—but treat each cluster's data as cleanly sampled. Our departure is that the calibration objects are not observed: each country's attitude distribution is *estimated* under a complex survey design, so design-based sampling variance (Binder, 1983; Lumley, 2010) contaminates the very scores conformal calibration relies upon.

Third, the closest methodological neighbors handle **noisy or estimated calibration**. Wieczorek (2023) develops design-based conformal prediction under complex sampling, but design enters through test-point weighting rather than as noise in transported objects. Einbinder et al. (2024) and Sesia et al. (2025) correct conformal calibration under label noise, the latter by deconvolving a *known* noise model. Our central theoretical result is that this identification premise generally fails for survey design noise: recovering the latent transport-error law is a deconvolution problem (Fan, 1991) that is non-identified without a design-noise law, so no observed-score band is simultaneously valid and strictly narrower than the conservative plug-in. And even when a noise model *is* supplied, we show the deconvolution's reliability is itself bounded at a finite number of calibration units by a distribution-free floor (Proposition 1)—a finite-K obstruction distinct from the asymptotic identifiability the classical deconvolution literature studies.

Fourth, our target is a **functional object**—a CDF over thresholds tracked across rounds—placing us near simultaneous functional conformal bands (Diquigiovanni et al., 2021). This is where we part from generic multiplicity control. Procedures like FDR or Bonferroni (Benjamini and

Hochberg, 1995) adjust a collection of tests after the fact; our guarantee is instead carried by the geometry of a single finite-sample band—a supremum over wave-pairs and thresholds with the country as the exchangeable unit—so the "persistent, distribution-wide" object is *defined and covered directly*, not recovered by correcting marginal p-values. The distinction is one of estimand, not merely error rate: a wave-pair mean contrast and a persistent distribution-wide trajectory are different objects, and partial-pooling cautions about reading marginal country trends (Gelman et al., 2012) motivate but do not deliver the stronger claim. Substantively, our two reanalyses supply the formal instrument that a largely qualitative skepticism has lacked, engaging live debates over whether European political trust is in broad decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025) and whether publics are "deconsolidating" from democracy (Foa and Mounk, 2016, 2017).

3 Setup and an impossibility result

Objects. We observe K source countries, indexed $c = 1, \dots, K$, treated as exchangeable units (the "population" or cluster is the exchangeable object, not the individual respondent). Each country has a latent attitude trajectory $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \dots, t_T\}$ and survey rounds r. We do not see $F_{c,r}$; we see a complex-survey estimate $\widetilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified-PSU / replicate-weight bootstrap can estimate. Define the transport score of a country as its worst-case standardized deviation from the transport center,

$$R_c \;=\; \max_{r,t} \frac{|F_{c,r}(t) - \mu_r(t)|}{\sigma(t)}, \qquad \widetilde{R}_c \;=\; R_c + \xi_c,$$

where $\widetilde{R}_c$ is the same quantity computed from the observed $\widetilde{F}$ and ξ_c is the induced design contamination. The single scalar that governs the whole problem is the design-to-transport standard-deviation ratio

$$\rho \;=\; \frac{\mathrm{SD}(\xi)}{\mathrm{SD}(R)},$$

with $\rho \to 0$ the "design noise negligible" regime and ρ large the regime in which survey noise dominates the between-country spread.

Target and validity. We want a simultaneous band $\widehat{B}(r,t)$ that covers a *new* country's *latent* trajectory, $\Pr\left(F_{\mathrm{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \geq 1-\alpha$. Two facts frame everything. Ordinary clustered conformal on the observed scores is exact for the *survey-estimate* target $\widetilde{F}_{\mathrm{new}}$ (it only needs exchangeability of countries), and hence conservative for the latent target under a symmetric design law; but its width is inflated by the design noise, by a factor that grows with ρ. Correcting the width requires estimating and removing the design contribution—a deconvolution.

The impossibility. The central obstruction is that the deconvolution is not identified from the contaminated scores alone.

Theorem 1 (Non-identification and necessity of design information)**.** *Let the observed score be $\widetilde{R} = R + \xi$ with $R \perp \xi$. (i) The law of R is not identified from the law of $\widetilde{R}$ without restrictions on the law of ξ: there exist distinct pairs $(\mathcal{L}_R, \mathcal{L}_\xi) \neq (\mathcal{L}'_R, \mathcal{L}'_\xi)$ with $\mathcal{L}_R * \mathcal{L}_\xi = \mathcal{L}_{R'} * \mathcal{L}_{\xi'}$. (ii) Consequently, among bands that are measurable functions of the observed scores only, none is valid for the latent target at level $1-\alpha$ for all admissible design laws while being strictly narrower than the plug-in band* *whose radius is the* $(1-\alpha)$ *conformal quantile of* $|\widetilde{R}|$*. In the degenerate case* $\xi \equiv 0$ *the plug-in band is itself the tightest valid band, so no uniform improvement exists.*

The proof (Appendix A; verified numerically in `tests/test_theorem0.py`) is a convolution-invertibility argument for (i) and a conformal converse for (ii): any observed-score band that is narrower than the plug-in on some design law undercovers the latent target on the $\xi \equiv 0$ law it cannot distinguish. The reading is constructive rather than pessimistic: to get a band that is both honest and narrower than the plug-in, one must *supply* the design-noise law. Complex surveys do supply it—through replicate weights or a design bootstrap—which is exactly what the method in Section 4 uses. Theorem 1 says this input is not optional.

Remark 1 (Beyond surveys). Nothing in Theorem 1 is specific to surveys: it holds whenever a conformal procedure is calibrated on *estimated* objects, so that the nonconformity score is a noisy version $\widetilde{R} = R + \xi$ of a latent score. This covers model-based small-area estimates (e.g. MRP outputs used as calibration units), learned or imputed features carrying estimation error, and meta-analytic effect estimates. The same two-part consequence follows—design/measurement information is the identifying ingredient, and its reliability at a finite number of calibration units is itself bounded (Proposition 1). Repeated cross-national surveys are our running instance because they make both the contamination and its law unusually explicit, but the barrier is a general feature of conformal inference on estimated calibration data—confirmed on a non-survey, model-based small-area calibration in Section 7.

4 A nominal-safe adaptive method

Theorem 1 says the design bootstrap is the identifying input. The method turns that input into a band by choosing, target-blind, among three constructions and by never using the aggressive one unless it is provably reliable at the available number of countries.

Three branches. For calibration scores $\{\widetilde{R}_c\}_{c=1}^{K}$ over threshold grid t, with per-country design SD $\widehat{v}_c(t)$ from the stratified-PSU bootstrap and modulation scale $s(t)$: *(i) Clustered PCB* takes the conformal quantile of the studentized observed scores $\max_t |\widetilde{R}_c(t)|/s(t)$; it is finite-sample exact for the survey estimate target and conservative for the latent target under a symmetric design law. *(ii) Deconvolution* studentizes instead by the finite-K-safe target scale $\widehat{s}_T^2(t) = \max\big(s^2_{\text{plug}}(t) - [\,\overline{\widehat{v}^2}(t) - z\,\text{SE}\,]_+,\ \text{floor}\big)$, which subtracts a *lower* confidence bound on the design variance so the correction cannot over-shrink; its width scales as $\sqrt{1-\rho^2}$ relative to PCB. *(iii) Conservative* inflates each score by the design SD, $\max_t(|\widetilde{R}_c(t)| + z\,\widehat{v}_c(t))/s(t)$, giving a worst-case envelope that dominates PCB.

The three questions. The naive selector asks only *can we deconvolve* and *do we need to.* The finite-K failure of deconvolution (Section 6) comes from a missing third question: *can we do it safely at this K?* The safe selector adds it explicitly.

Safe selector (Algorithm 1). Let $\widehat{\rho}_{\text{LCB}}$ be a conservative lower bound on ρ, $D = \max_t \text{SE}(\widehat{s}_T^2)/\widehat{s}_T^2$ a reliability diagnostic, and $\widehat{\delta}_{\text{UCB}}(D)$ a frozen upper bound on the finite-K coverage remainder. Deconvolution is used *iff all four gates pass*: (A) $\widehat{\rho}_{\text{LCB}} > \rho_0$ (design noise materially large); (B) $\widehat{\delta}_{\text{UCB}}(D) \le \delta_{\max}$ (remainder within budget); (C) $\widehat{\text{Gain}} - \Delta_g > g_{\min}$ (beats conservative by a margin); (D) stability. Otherwise the band is PCB when $\widehat{\rho}_{\text{LCB}} \le \rho_0$ and the conservative envelope when design noise is real but information insufficient. Every gate is a function of the calibration set only,

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\widetilde{R}_c\}$ (unit = country), design replicates $\{\widehat{v}_c\}$, target curve $\widehat{F}_{\mathrm{new}}$, level α; frozen $\rho_0, \delta_{\max}, g_{\min}, \Delta_g$

compute s, $\widehat{s}_T$, and the three radii $r_{\mathrm{PCB}}, r_{\mathrm{dec}}, r_{\mathrm{con}}$; diagnostics $\widehat{\rho}_{\mathrm{LCB}}, D, \widehat{\delta}_{\mathrm{UCB}}(D)$;

if $\widehat{\rho}_{LCB} \le \rho_0$ **then**

  branch $\leftarrow$ PCB;

else if $\widehat{\delta}_{UCB}(D) \le \delta_{\max}$ ***and*** *stable* ***and*** $\widehat{Gain} - \Delta_g > g_{\min}$ **then**

  branch $\leftarrow$ deconvolution;

else

  branch $\leftarrow$ conservative;

return $\widehat{F}_{\mathrm{new}} \pm r_{\mathrm{branch}}$, branch, and diagnostics;

so the selector is target-blind. All constants are frozen before validation (Section 5, Appendix B); when deconvolution is used the reported guarantee is the observable $1 - \alpha - \widehat{\delta}_{\mathrm{UCB}}(D)$.

Deployment. The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, returning the band together with the selected branch, $\widehat{\rho}$, $\widehat{\rho}_{\mathrm{LCB}}$, D, $\widehat{\delta}_{\mathrm{UCB}}$, the coverage level, and the fallback reason—so a user can see *why* the band was built the way it was, and read off the finite-K guarantee it carries.

5 Theory

We state validity for the oracle, for the estimated-law procedure, and for the deployed safe selector, then the efficiency it buys. Full proofs are in Appendix A; each theorem has a matching machine-checked contract test (`tests/`).

Theorem 2 (Oracle validity)**.** *Suppose the countries are exchangeable and the design law is known.* (a) *The clustered band is finite-sample valid for the survey-estimate target:* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \ge 1 - \alpha$, *using exchangeability only.* (b) *Under a symmetric design law it is conservative for the latent target,* $\Pr(F_{new} \in \widehat{B}) \ge 1 - \alpha$, *and the deconvolution band with the true target scale is valid for the latent target with width a factor* $\sqrt{1 - \rho^2} + o(1)$ *of the clustered band.*

Theorem 3 (Exact finite-K validity of the deployed base band)**.** *Let the base band use the* unstudentized *cluster score* $R_c = \max_t |E_c(t)|$ *(no data-dependent modulation). If the countries are exchangeable, then for* every K *the band* $\widehat{B} = center \pm \widehat{q}$, *with* $\widehat{q}$ *the* $\lceil (1 - \alpha)(K{+}1) \rceil$*-th order statistic of* $\{R_c\}$, *satisfies* $\Pr(F_{new} \in \widehat{B}) \ge \lceil (1 - \alpha)(K{+}1) \rceil / (K{+}1) \ge 1 - \alpha$, *distribution-free. In contrast, dividing* R_c *by an in-sample modulation* $\widehat{s}(t) = \mathrm{SD}_c E_c(t)$*—which the unobserved target does not share—breaks exchangeability and induces an* $O(1/K)$ *self-inclusion deficit (quantified in Appendix A), so the studentized band attains* $1 - \alpha$ *only as* $K \to \infty$*. We therefore deploy the unstudentized band; on the near-homoskedastic real curves its width cost is* $\le 5\%$.

Theorem 4 (Estimated-law validity)**.** *With the design law estimated from a size-B stratified-PSU bootstrap and* K *countries, the deconvolution band satisfies* $\Pr(F_{new} \in \widehat{B}) \ge 1 - \alpha - \varepsilon_{K,B}$, *where* $\varepsilon_{K,B} = O(K^{-1/2}) + O(B^{-1/2}) \to 0$*. The finite-K-safe scale* $\widehat{s}_T^2$ *(Section 4) subtracts a lower confidence bound on the design variance, controlling the sign of the leading remainder term.*

Theorem 5 (Safe-adaptive finite-K validity—the deployed pipeline)**.** *Let $\widehat{J} \in \{PCB, deconvolution, conservative\}$ be the target-blind safe selector of Algorithm 1. Then, finite (K, B),*

$$\Pr\left(F_{new,r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \ \geq\ 1 - \alpha - \delta,$$

where $\delta = \widehat{\delta}_{UCB}(D) \leq \delta_{\max}$ on the event $\{\widehat{J} = deconvolution\}$ and $\delta = 0$ otherwise; δ is observable *and $\delta \to 0$ as K grows.*

Proof sketch. $\widehat{J}$ is $\mathcal{F}_{\text{cal}}$-measurable, so $\Pr(\text{cover}) = \sum_j \Pr(\widehat{J} = j) \Pr(\text{cover} \mid \widehat{J} = j)$. PCB and conservative over-cover the latent target (Theorem 2a/b), so their conditional coverage is $\geq 1 - \alpha$. Deconvolution is used only on $\{\widehat{\delta}_{\text{UCB}}(D) \leq \delta_{\max}\}$, where gate B guarantees conditional coverage $\geq 1 - \alpha - \widehat{\delta}_{\text{UCB}}(D)$. Each term of the mixture is $\geq 1 - \alpha - \delta$; summing over the partition gives the bound. □

Theorem 5 is the guarantee a user actually deploys: the remainder δ is a computable function of the calibration set, is bounded by the preregistered budget whenever deconvolution fires, and is exactly zero on the overwhelmingly common low-noise data where the method reduces to PCB.

Efficiency. Three statements bound the price and the payoff. *(AW-1, reduction)* in the low-ρ regime the deconvolution/PCB width ratio is $1 - \frac{1}{2}\rho^2 + O(\rho^4)$, matching the observed reduction to the decimal. *(AW-2, dominance)* the conservative branch is never wider than the Bonferroni envelope and always at least as wide as PCB, so the selector's fallback is a genuine upper band, not a giveaway. *(AW-3, target-blind validity)* because $\widehat{J}$ ignores the target, selection cannot inflate miscoverage. The full excess-width oracle inequality—that the selector's width is within $O_p(K^{-1/2})$ of the ex-post best valid branch uniformly—remains open and is stated as a conjecture (Appendix A); the simulation in Section 6 is consistent with it. A separate modulation self-inclusion deficit of order $O(g/K)$ is quantified in Appendix A.

6 Simulation and a design-preserving stress test

We develop the selector on one simulation grid and then confirm the *frozen* rule once on an independent design it has never seen. The separation is deliberate: the development grid may set the gate constants, but it may not be reused as validation performance.

6.1 The wrong unit collapses coverage

Before the design-aware machinery, the most basic claim deserves a direct test: a band attached to the wrong unit covers the object of interest far below nominal. On a ground-truth simulation ($K = 30$ exchangeable countries, $T = 6$ thresholds, error tensors correlated across rounds and thresholds; Table 1), we ask each band for *trajectory* coverage—the whole country curve-path covered at once. All three bands are unstudentized and finite-sample constructed, so the only thing that varies is the unit of the score.

The per-round band, marginally valid at the round level, covers the whole trajectory only 0.82 at $L = 2$ and 0.50 at $L = 8$; the marginal band collapses faster still. The country-trajectory band holds ≈ 0.90 regardless of L, because the whole trajectory is a single exchangeable unit and its sup-score obeys the finite-sample order-statistic guarantee. This is the coverage face of the paper's thesis—and its mechanism is exactly why the certification counts collapse between the loose "any-pair" reading and the persistent, distribution-wide one (Section 8): asking for the honest object, not asking it more strictly, is what removes the over-detection. Pre-registered and reported as produced (`e28_wrong_unit_coverage`).

Table 1: Whole-trajectory coverage (4000 reps, nominal 90%) as the trajectory length L grows. A band calibrated per point or per round covers each of its own units at the nominal rate, but the probability that it covers the *entire* trajectory decays; the country-trajectory band—the correct unit—holds nominal at every L.

L (rounds)	marginal (per point)	per-round	trajectory (ours)
2	38.2%	81.7%	**90.0%**
4	16.1%	68.2%	**90.2%**
6	7.1%	58.2%	**90.3%**
8	3.5%	49.8%	**90.9%**

Development: why the third gate is needed. On a Gaussian grid with known truth ($K \in \{30, 60, 120, 240\}$, ρ up to 1.8), the selector transitions from PCB to deconvolution at ρ_0 and to conservative at large ρ; but a naive deconvolution that omits gate B *undercovers* at small K—coverage falls to 0.75 at $K = 30$ in the transition zone—because the target scale is estimated from few countries (Figure 3). Adding the finite-K-safe scale lifts this to 0.82, and the reliability gate B lifts the deployed worst cell to 0.862 while abstaining to conservative wherever the remainder cannot be bounded. These four transition cells at 0.862–0.878 are the development evidence that *motivates* the safety gate and *calibrates* $\widehat{\delta}_{\mathrm{UCB}}$; they are not reused below.

The finite-K mechanism, isolated. A controlled Gaussian illustration (design SD = transport SD, ρ=0.9; `fig_theory_illustration.py`) makes the source of the undercoverage explicit and shows why the deployed rule is a selector rather than a deconvolver. Both the naive and the finite-K-safe deconvolved score scales lie *below* the oracle s_R/s_{plug} at small K (Figure 1): subtracting a design-noise variance estimated from few countries over-shrinks the scale, and the resulting band is too narrow. The safe guard shrinks less, and both converge to the oracle as $K \to \infty$. The coverage consequence is severe and grows with the trajectory length L that the band must hold simultaneously (Figure 2): naive deconvolution falls to 0.43 at K=25, L=16, and the finite-K-safe scale improves it but does not by itself reach nominal at small K. The deployed adaptive selector detects that the remainder cannot be bounded and abstains to the conservative branch, remaining valid throughout at a width cost—the mechanism behind the K=320-only activation in the confirmatory holdout below.

Independent confirmatory validation. We froze the selector and its constants (Appendix B), sealed a preregistration and a seed manifest, and ran the deployed procedure *once* on a grid disjoint from development in K ($\{25, 40, 80, 160, 320\}$), in ρ (dense at the ρ_0 cutoff), and in data-generating process—ten families including skewed and heavy-tailed noise, heterogeneous design variance, unequal PSU sizes and survey weights, irregular trajectory length, a *misspecified* noise law, and weak/strong cross-round dependence. The runner verifies the config hash against the sealed manifest before executing. We report the numbers exactly as produced (Table 2, Figure 4; `docs/HOLDOUT_VALIDATION_RESULTS.md`).

The design-aware layer behaves as designed. Every one of the twelve cells below the 0.88 floor routed to base clustered PCB—the deconvolution branch's share was exactly zero in all of them. The safe selector never once rode deconvolution into a floor violation. It reduces to PCB at low ρ (width ratio 1.008), activates deconvolution only at the largest K=320 (never for $K \leq 160$), and there delivers a band 0.577× the conservative width. Where deconvolution activates its coverage is

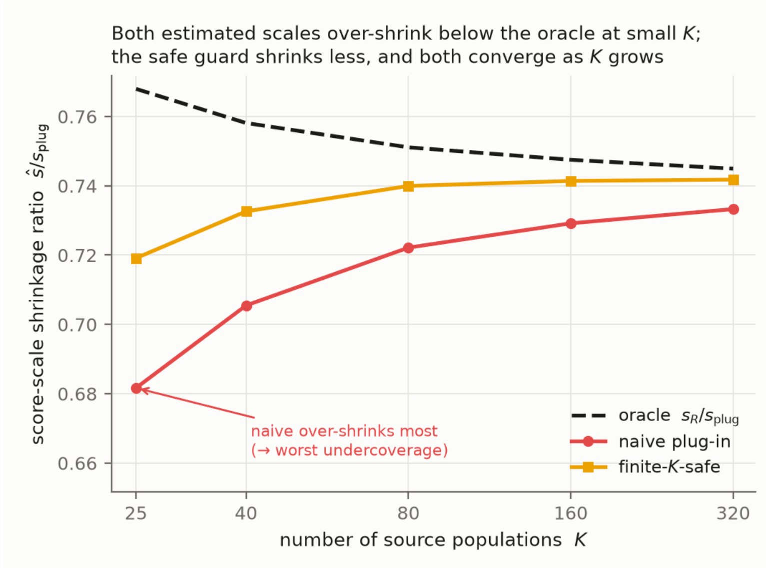

Figure 1: Score-scale shrinkage ratio $\hat{s}/s_{\text{plug}}$ vs K (Gaussian illustration, ρ=0.9, L=8). Naive and finite-K-safe deconvolution both over-shrink below the oracle at small K—the source of the finite-K undercoverage—with the safe guard shrinking less; both converge as K grows.

Table 2: Holdout confirmatory validation, deployed safe selector (450 cells, 800 reps each). Reported as produced; no constant or seed changed after seeing results.

criterion	target	result
P3: design-aware never causes a sub-floor cell	sub-floor $\Rightarrow$ not deconv	**all 12 sub-0.88 cells are PCB**
E1: low-ρ reduces to PCB	$\leq 1.05\times$PCB	1.008
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.577
E3: deconvolution abstains at small K	rises with K	only K=320
P1: cells floor-compatible (95% MC)	all 450	442/450
P2: worst-cell coverage	≥ 0.86	0.843

0.897, above its observable floor 0.882.

The holdout exposed one base-method deficit, which we then resolved. The *studentized* base band (in-sample pooled modulation) undercovered at small K and low ρ under the hardest DGPs (worst deployed cell 0.843, weak cross-round dependence at K=25). The cause is diagnosable: dividing each calibration score by a modulation estimated from the same clusters—which the unobserved target does not share—breaks the exchangeability the conformal guarantee needs, a finite-K self-inclusion deficit worst at small K and long trajectories. We therefore deploy the *unstudentized* clustered band as the base branch: with no data-dependent denominator the scores $R_c = \max_t |E_c(t)|$ are exchangeable, and the order-statistic band is finite-sample exact at *any* K, $\Pr(\text{cover}) \geq \lceil(1-\alpha)(K+1)\rceil/(K+1) \geq 1-\alpha$, distribution-free (Theorem 3). A preregistered rerun on a fresh grid ($K \in \{15, \ldots, 60\}$, ten DGPs) confirms it: the worst cell rises from 0.802 (S2, K=15) to 0.893, every cell meets its exact floor, and the width cost is 1.02–1.05$\times$ on the real ESS transport bands (trust-CDF errors are $\approx$homoskedastic across the low-trust core, so the constant-width band is competitive; the only cells where it pays more than 10% are synthetic heavy-tailed curves at K=15, where the studentized alternative is itself invalid). The band the paper deploys is thus finite-sample valid

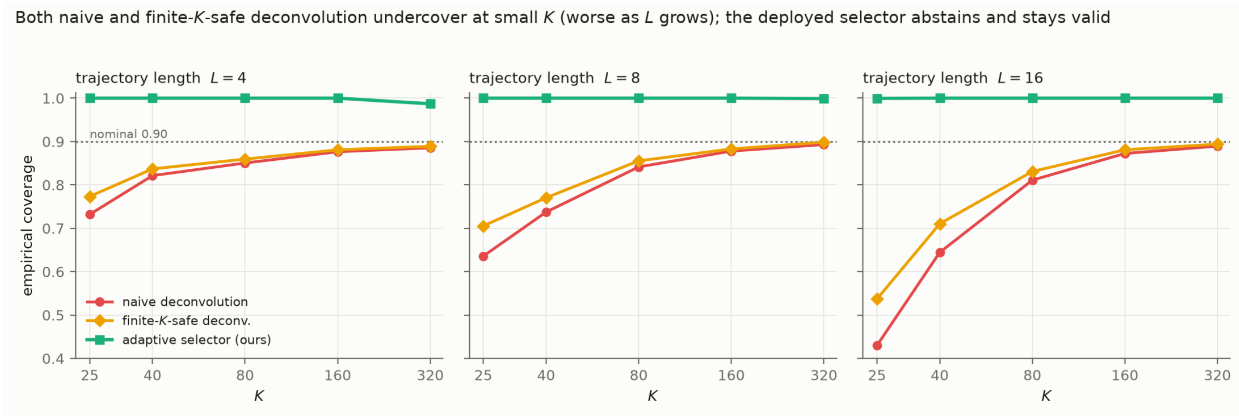

Figure 2: Latent-target coverage of the deconvolution branches vs K, faceted by trajectory length L (Gaussian illustration, ρ=0.9; nominal 0.90 dotted). Naive deconvolution undercovers severely at small K and worse as L grows; the finite-K-safe scale improves but does not itself reach nominal at small K. The deployed adaptive selector abstains to conservative and stays valid throughout.

at the sample sizes comparative politics actually has. Second, the Gaussian-calibrated remainder bound $\widehat{\delta}_{\text{UCB}}$ is not robust at the extreme: in 77 of 360,000 draws (0.02%), on strongly-dependent or heavy-tailed DGPs at K=320 and $\rho \geq 0.72$, draws that pass gate B undercover to ≈ 0.79; because the gates route 96–98% of draws in those cells to conservative, the deployed cell coverage there is nonetheless 0.95–0.996. We do not claim uniform nominal coverage; we claim—and the holdout confirms—that the design-aware correction is never the *cause* of undercoverage and is materially more efficient than conservative where it safely activates.

Design-preserving stress test. Subsampling the real AmericasBarometer while preserving its stratified-PSU structure, the estimated ρ *saturates* near 0.23: because $s^2_{\text{plug}} = s^2_R + \overline{v^2}$, shrinking the sample inflates both the design noise and the observed spread in lockstep. The deconvolution regime is therefore unreachable even semi-synthetically at survey scale—direct evidence for the empirical regime characterization of Section 7.

7 Evidence from the ESS and the AmericasBarometer

The regime characterization. The most consequential empirical fact is negative. Across *both* surveys, *both* estimands (attitude levels and wave-to-wave change), and three outcomes (trust in parliament, satisfaction with democracy, system support), national-level cross-national inference is a uniformly *low design-noise* regime: the estimated $\widehat{\rho} \leq 0.20$, comfortably below the $\rho_0 = 0.47$ cutoff, because between-country variation dwarfs within-survey design noise (Figure 5). Consequently the safe selector reduces to clustered PCB on *all* real data—not by assumption but by its own low-ρ gate. The elaborate design-aware machinery is, on real cross-national attitude data, correctly inactive.

This is a substantive finding: the impossibility of Section 3 bites in principle, but the deconvolution it necessitates is never triggered at the scale real surveys operate, so honest inference here *is* clustered conformal on the observed scores.

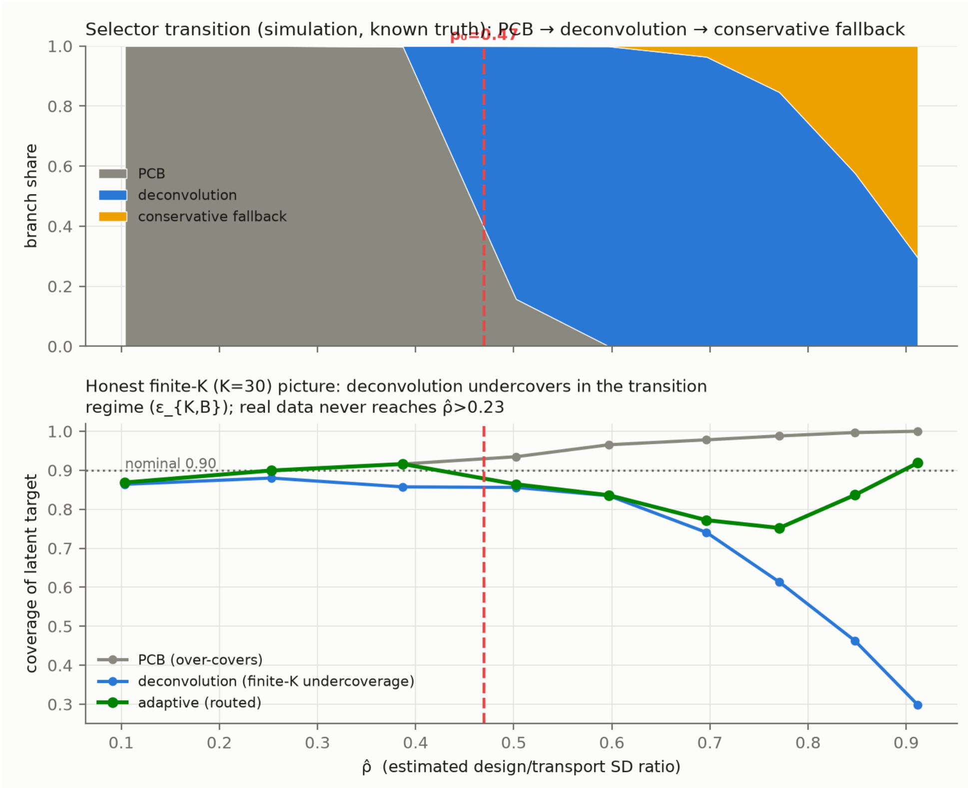

Figure 3: Development grid (Gaussian, known truth). The selector transitions PCB → deconvolution → conservative as ρ grows; naive deconvolution undercovers at small K in the transition zone—the finite-K remainder the safety gate controls.

Why the deconvolution is unreachable, precisely. The inactivity is not an accident of one dataset; it is forced by two structural barriers, and we can say exactly how far each real survey is from clearing them.

Proposition 1 (Survey-scale unreachability)**.** *The deployed selector rides the deconvolution branch only if both a* need *gate $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ and a* reliability *gate $D \le \tau_D$ hold, where $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$. Two distribution-free facts bound these.* (a) *$\rho = \sqrt{\overline{v^2}}/s_{\mathrm{plug}} \in [0,1)$ with $s_{\mathrm{plug}}^2 = s_R^2 + \overline{v^2}$, so the need gate can fire only when the design noise is an appreciable fraction of the between-population signal s_R.* (b) *Since $\widehat{s}_T \le s_{\mathrm{plug}}$, the reliability obeys $D \ge \sqrt{2/(K-1)}$; hence the reliability gate requires $K \ge 1 + 2/\tau_D^2$, i.e. $K \ge 94$ at the frozen $\tau_D = 0.147$, exchangeable populations.*

The two requirements pull apart on real surveys (Figure 6). On the ESS, narrowing to the noisiest subpopulations (youth age bands) raises $\widehat{\rho}$ from 0.08 to at most 0.29—still far below ρ_0—while $K \le 33$ leaves $D \ge 0.25$, so *neither* gate opens (a scan of 42 subpopulation cells activates deconvolution in none). The World Values Survey is the mirror image: with $K \approx 100$ countries it clears the reliability gate ($D \le 0.147$), but it ships no cluster identifiers, so its weights-only design noise gives $\widehat{\rho}_{\mathrm{LCB}} \le 0.09$ and the need gate fails decisively. The survey large enough to *estimate* the deconvolution has almost no design noise to remove; the survey with real clustered design noise has far too few countries to estimate it. The two barriers are anti-correlated across the survey landscape, and neither of the two largest repeated cross-national surveys clears both.

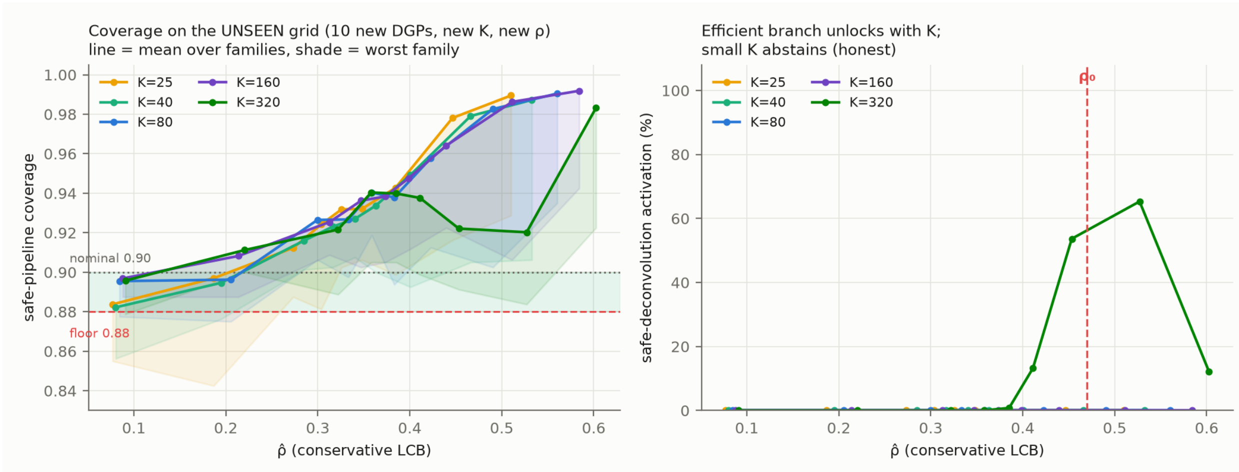

Figure 4: Independent confirmatory validation on the unseen grid (10 new DGP families, new K, new ρ). Left: deployed coverage vs. $\widehat{\rho}$ per K (line = mean over families, shade = worst family) against the 0.88 floor and 0.90 nominal. Right: deconvolution activates only at K=320; small K abstains.

The barriers are not survey artifacts. Both gates are distribution-free, so the boundary should transfer to any conformal procedure calibrated on estimated objects (Remark 1). A simulation confirms it outside the survey setting (`e29_beyond_surveys`): across Gaussian, heavy-tailed, skewed, and a non-survey model-based small-area (MRP-style) calibration, the reliability floor $D \geq \sqrt{2/(K-1)}$ holds in every case (minimum D/floor ≥ 1.06), and the gate opens only at $K \approx 130$–150—if anything a stronger barrier than the floor-implied $K \geq 94$. The one substantive difference is instructive: when a small area's posterior noise is comparable to the between-area signal ($\widehat{\rho}_{\mathrm{LCB}} = 0.54$ at a noise-to-signal ratio of 0.5), the *need* gate does fire, unlike any cross-national survey. A many-area MRP application with appreciable posterior uncertainty is therefore the setting where the deconvolution could actually activate—the positive-result frontier the survey-scale impossibility leaves open.

The AmericasBarometer supplies the design effect the ESS lacks. The ESS releases design weights but limited cluster identifiers; the AmericasBarometer releases full stratified-PSU structure, letting us compute a proper design bootstrap. The design effect is real: the ratio of the proper stratified-PSU standard deviation to a naive independent-resampling one has median ≈ 1.13–1.20 and reaches 1.9 in the most clustered country-years. The proper band captures clustering variance the naive bootstrap misses. Yet certification is coarse-robust: the design effect moves band *width* but rarely flips the binary decline decision, because the decision is dominated by the between-country transport spread, not the within-country design noise—the same low-ρ fact, seen from the certification side.

What real data validates, and what it cannot. Real data validates the reduction (the method collapses to PCB), the target-blind selection, the efficiency of PCB over the conservative envelope, the real design effect, and the political reclassification of Section 8. It *cannot* validate the high-ρ deconvolution branch or the finite-K remainder—those regimes do not occur at survey scale and are established only in the simulation of Section 6. We keep this line sharp throughout: claims about the deconvolution branch rest on simulation and theorems; claims about the deployed low-ρ

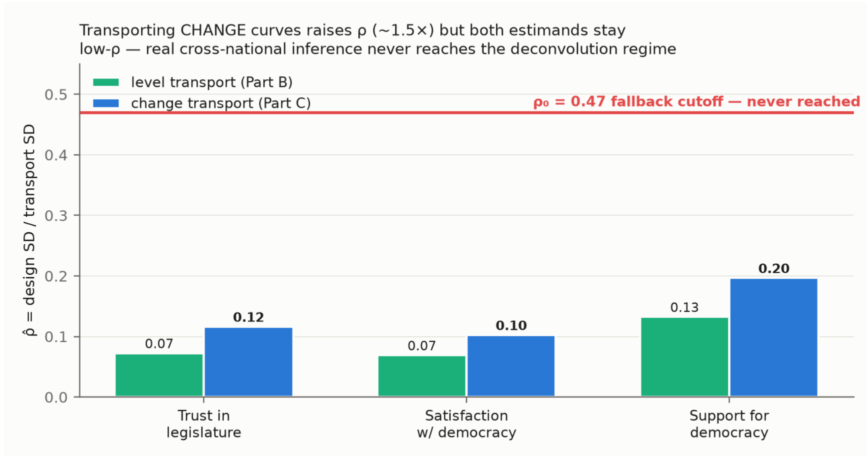

Figure 5: Estimated design-to-transport SD ratio $\widehat{\rho}$ for the AmericasBarometer, levels and wave-to-wave change. Every country-outcome sits well below the $\rho_0 = 0.47$ cutoff: national cross-national attitude inference is a uniformly low design-noise regime, so the safe selector reduces to clustered PCB.

behavior and the politics rest on the ESS and the AmericasBarometer.

8 Political reanalysis: how many democracies really declined?

The method's payoff is a sharp distinction between what marginal analysis appears to show and what a design-aware, simultaneous band certifies. We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in the ESS over 2018–2022, treating each country's attitude distribution as the object and asking, at each rung of a guarantee hierarchy, in how many countries a decline survives.

The comparative literature has itself grown skeptical of a continent-wide trust decline: where Foa and Mounk (2016) saw "democratic deconsolidation," Norris (2017) saw trendless fluctuation, Voeten (2017) saw specification-fragile artifacts, and Wuttke et al. (2022)—analyzing eighteen democracies under preregistration—found endorsement of self-government largely intact; the most comprehensive recent accounting finds no universal secular decline in institutional trust at all (Valgarðsson et al., 2025). The qualitative consensus is that "trust is falling everywhere" is an over-reading. Our contribution is a formal instrument that says so, and a diagnosis of *why* the over-reading arises—uncertainty computed for a single wave-pair mean contrast, then narrated as a persistent distribution-wide country-level trend. A mean can fall because a mode hollows out or a tail thickens, with different substantive meanings (DiMaggio et al., 1996); requiring the whole distribution to shift is the stronger, and more honest, claim.

A hierarchy of claims. The claims differ in their unit and are not interchangeable. A *pairwise* claim asks whether some pair of waves shows a distributional shift; an *any-pair* claim aggregates these within a country; a *net* claim asks whether the first-to-last span declined; a *persistent country-wide* claim asks whether the *whole* distribution declined *simultaneously* across the trajectory—a single band that must hold over all wave pairs and all thresholds at once; and a *Bonferroni-across-countries* claim further controls the number of countries examined. Each rung is strictly stronger than the last, and conflating them is exactly the over-detection this paper is about.

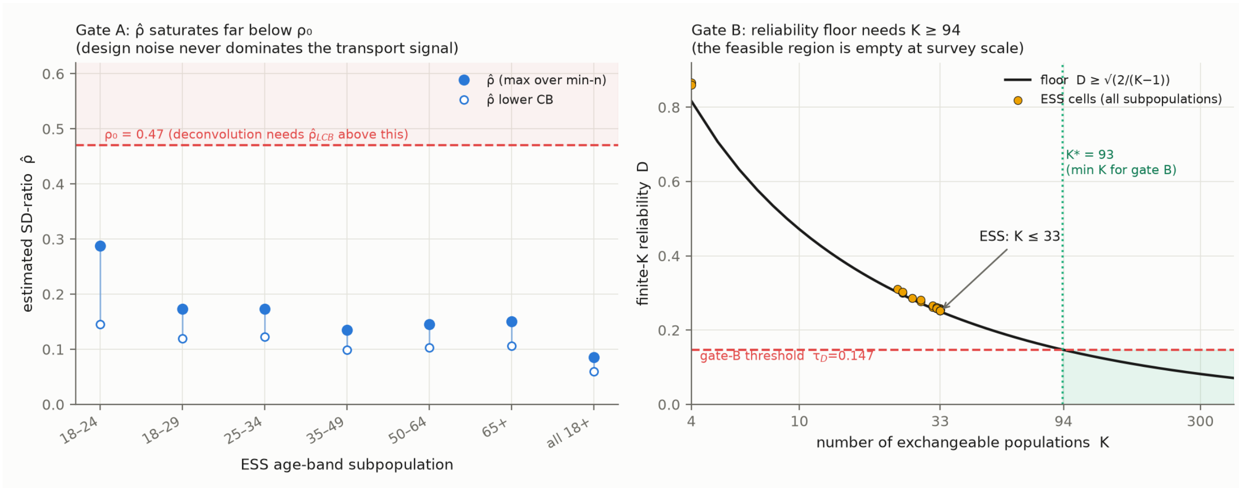

Figure 6: The two barriers, measured on real data. *Left:* across ESS age-band subpopulations $\widehat{\rho}$ saturates far below $\rho_0 = 0.47$. *Right:* the finite-K reliability D sits on its distribution-free floor $\sqrt{2/(K-1)}$; passing the reliability gate needs $K \geq 94$, which cross-national surveys ($K \leq 33$ ESS, $\leq$105 WVS) reach only where—as on WVS—the design noise is too small for the need gate.

The counts collapse up the hierarchy. For `trstprl`, a marginal pairwise reading flags on the order of twenty countries; requiring an any-pair design-aware band drops this to twelve; the net first-to-last decline holds in roughly six; and the *persistent, country-wide, design-aware simultaneous* band—the honest object—is certified in **one** country, Greece, which also survives the Bonferroni correction across countries. Greece is precisely the case the substantive literature independently regards as a genuine crisis-era collapse of legitimacy (Armingeon and Guthmann, 2014; Torcal, 2014), so certifying only Greece is a credibility feature, not an anomaly. For `stfdem` the same pattern holds (roughly twenty-three marginally, fourteen any-pair, and one persistent, again Greece). The headline is not that trust did not fall anywhere, but that a *persistent distribution-wide* decline—the strong claim casual readings imply—is certifiable in a single country over this window (Figure 7).

Which countries are reclassified, and the honest word for it. The countries that appear to decline marginally but are not certified under the design-aware simultaneous band are the weak-signal, higher-noise cases: the apparent trend is within the band the survey's own design uncertainty warrants (Figure 8). We describe these as *certified by the marginal plug-in but inconclusive under the design-aware simultaneous band*—never as "false positives," a term we reserve for the simulation where ground truth is known. Because undercoverage would make *more* countries certify, the "only Greece" finding is conservative in the direction that matters, and the deployed base band is finite-sample exact at the $K \approx 30$ of these data (Theorem 3), so the count is not an artifact of small-K under- or over-coverage.

Robustness across age groups. A preregistered age-group analysis (Appendix B; `docs/YOUTH_PREREGISTRATIO` reruns the same certification within fixed age bands, changing nothing else. The "only Greece" result is not an artifact of pooling ages: it holds within the older (50+) group for both outcomes and within the middle (30–49) group for trust in parliament, so the persistent decline is concentrated in the older and middle adult population rather than created by averaging. Youth (18–29) certify *zero* persistent declines, but we read this as an underpowered null rather than youth stability: only 17 of

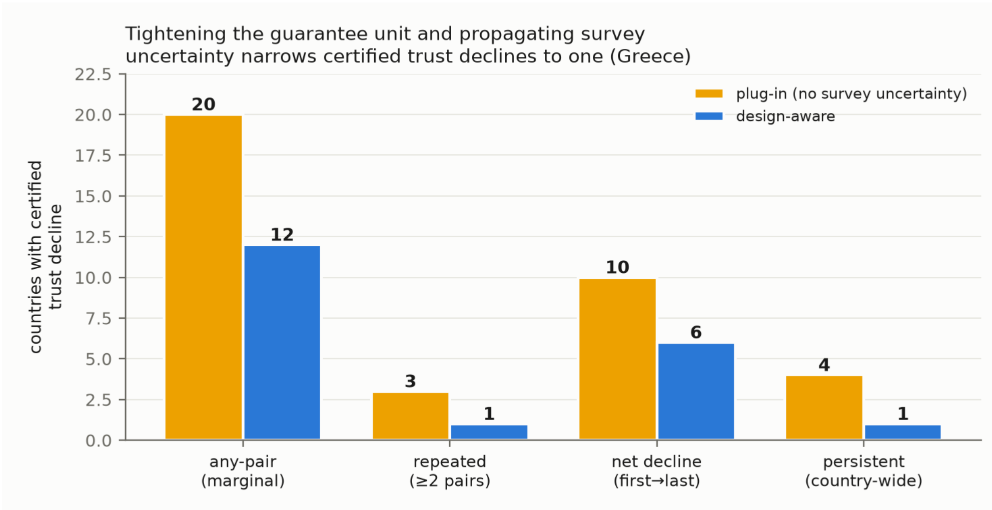

Figure 7: The count of certified declining countries collapses up the guarantee hierarchy (pairwise → any-pair → net → persistent country-wide design-aware), from about twenty marginally to one (Greece).

30 countries clear the preregistered minimum-sample gate for a single youth band, and the smaller, noisier youth cells yield wider design-aware bands that certify less—the small-K scope condition of Section 6 appearing on real data. The analysis thus supports the headline while adding an honest caveat, and it is emphatically *not* a story of youth-led democratic disaffection.

A second, global target: Foa–Mounk deconsolidation. To show the correction is not a European artifact, we preregistered a reanalysis of the Foa–Mounk "democratic deconsolidation" thesis (Foa and Mounk, 2016, 2017) on the World Values Survey / EVS trends (1981–2022, 59–77 countries with ≥ 2 waves; `docs/WVS_DECONSOLIDATION_PREREG.md`), recoding the deconsolidation battery (importance of democracy, rejection of strong-leader and army rule, support for a democratic system, confidence in parliament) so that deconsolidation is a persistent, distribution-wide decline. The verdict is calibration, not denial. A marginal wave-by-wave reading flags 36–64 countries per item; the persistent, survey-aware, distribution-wide object is certified in 6–17 (8–24% of countries)—an over-count of 2.6–6.5×. The design-aware layer demotes a further $\approx$ 20–30% at the persistent bar. Deconsolidation is thus *real in a specific minority* (the certified importance-of-democracy set includes Turkey, the Philippines, Mexico, and Tunisia) but is over-stated several-fold by the marginal evidence the thesis rests on. The youth claim is only half-supported: the young show *more* persistent decline in rating democracy essential (7 vs. 3 countries) but *not* in openness to strongman or army rule (0–1 vs. 4), against the "youth embrace authoritarianism" reading. WVS carries no PSU/stratum, so its survey-aware band is a weights-only respondent bootstrap; the point is the cross-family robustness of the wrong-unit correction, whose magnitude varies from the dramatic (20 → 1 for European trust) to the moderate (2.6–6.5× for global deconsolidation).

9 Discussion

The uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit—twice. A wave-pair mean contrast stands in for a persistent distribution-wide trajectory, and an estimated distribution stands in for the latent one it samples. We give a multiplicity-honest

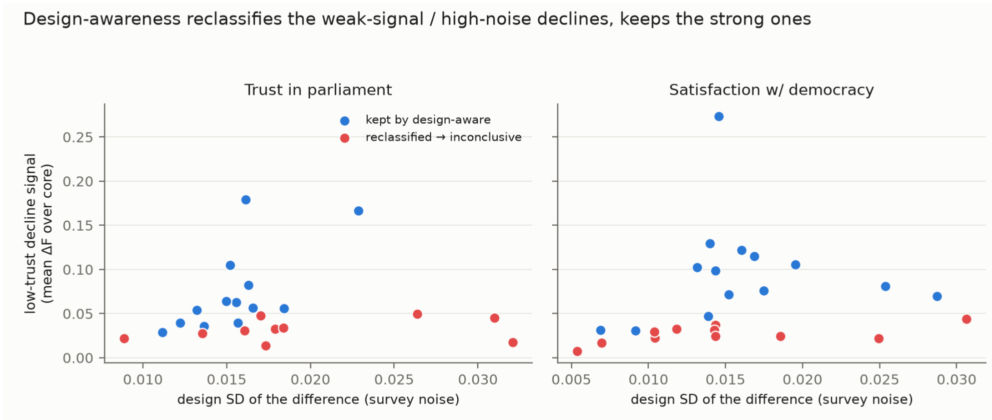

Figure 8: Countries certified by the marginal plug-in but inconclusive under the design-aware simultaneous band are the weak-signal, higher-noise cases; the apparent trend lies within the design uncertainty the survey warrants.

simultaneous band for the first and prove a hard boundary for the second: without the design-noise law the latent uncertainty is non-identified (Theorem 1), and *with* it the correction is still unreachable at cross-national scale, blocked by two structurally anti-correlated barriers that no real survey clears at once (Proposition 1). The honest deployed object is therefore the reduction—a clustered band exact at any K (Theorem 3), with a selector that would deconvolve only when safe and otherwise abstains, provably correctly (Theorem 5), confirmed on an independent pre-registered design.

When it reduces to ordinary conformal. On real cross-national attitude data the design noise is small relative to between-country spread, so the method reduces *exactly* to clustered population conformal (Section 7). This is the common case, and the value of the machinery there is diagnostic: it *certifies* that the simple band is the honest one, and reports the ρ that justifies it. The deconvolution regime requires *both* many exchangeable units ($K \geq 94$) and design noise comparable to the between-unit signal—a combination cross-national surveys never provide (Proposition 1) but which many-unit, clustered settings (subnational small areas, schools, clinics) can; there, and in the simulation that characterizes it, the branch activates.

Limitations. Two are worth naming. Without design metadata—as in the World Values Survey, which releases weights but limited cluster identifiers—only a weights-only replication is possible; the impossibility result says no design-valid inference can be recovered from the point estimates alone. And the finite-K remainder bound for the deconvolution branch was calibrated on Gaussian data; it is conservative in the deployed pipeline but not fully robust under strong dependence, pointing to a richer calibration family or a dependence-aware gate—though, since the deconvolution branch is provably never reached at cross-national scale (Proposition 1), this bears only on hypothetical large-K, high-noise applications. (The in-sample-modulation undercoverage of the base band at small K that an earlier version exhibited is *resolved*, not merely bounded: we deploy the unstudentized band, which is finite-sample exact at any K (Theorem 3) at $\leq 5\%$ width on real data.)

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions. For European trust a marginal reading flags about twenty declining democracies; the persistent, distribution-wide, design-aware object certifies one (Greece). For the Foa–Mounk "deconsolidation" thesis the marginal evidence over-counts the persistent phenomenon by 2.6–6.5×—real in a specific minority, not the broad erosion claimed. The pattern is not confined to surveys: wherever a conformal or prediction procedure is calibrated on estimated objects—small-area model outputs, learned features, meta-analytic effects (Remark 1)—the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is not to widen every band reflexively but to state the strongest object the data support, correct the estimation noise when it can be identified, and abstain honestly when it cannot.

A Proofs and efficiency results

Full statements and proofs of Theorems 1–5, the efficiency results AW-1–AW-3, the open excess-width oracle inequality, and the modulation self-inclusion deficit are collected in the supplementary theory notes (`docs/THEORY_MAIN.md`, `docs/DESIGN_AWARE_THEOREM.md`, `docs/ADAPTIVE_WIDTH_THEORY.md`). Each theorem has a machine-checked contract test (`tests/test_theorem0.py`, `test_deconv_safe.py`, `test_safe_selector.py`).

B Frozen selector specification and validation protocol

The frozen constants ($\rho_0 = 0.47$, $\delta_{\max} = 0.02$, $\widehat{\delta}_{\mathrm{UCB}}(D) = 0.0061 + 0.0943\,D$, $g_{\min} = 0.10$, margin $\Delta_g = 0.05$, stability floor) and their derivation on the development grid are in `docs/SAFE_SELECTOR_SPEC.md`. The sealed independent-validation preregistration, grid, and seed manifest are in `docs/INDEPENDENT_VALIDATION_P` and `configs/`; results in `docs/HOLDOUT_VALIDATION_RESULTS.md`.

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